

## Multiple Choice (10 marks)

1. Which of the following decimal numbers has an exact representation in binary notation?
  - (a) 0.1
  - (b) 0.2
  - (c) 0.3
  - (d) 0.5
2. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
  - (a) Recursive procedures
  - (b) Arbitrary goto's
  - (c) Two-dimensional arrays
  - (d) Integer-valued functions
3. A 3-way, set-associative cache is
  - (a) one in which each main memory word can be stored at any of 3 cache locations
  - (b) effective only if 3 or fewer processes are running alternately on the processor
  - (c) possible only with write-back
  - (d) faster to access than a direct-mapped cache
4. Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true of Church's thesis?
  - (a) It was first proven by Alan Turing.
  - (b) It has not yet been proven, but finding a proof is a subject of active research.
  - (c) It can never be proven.
  - (d) It is now in doubt because of the advent of parallel computers.
5. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
  - (a) Insertion sort
  - (b) Quicksort
  - (c) Merge sort
  - (d) Selection sort

## True / False (10 marks)

*Answer True or False*

6. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately  $1/K$ .
7. True or False: The addition of the two's-complement numbers 10000001 and 10101010 results in overflow in an 8-bit representation.
8. True or False: Local caching of files in distributed file systems always results in increased network traffic due to frequent data synchronization.
9. True or False: The one-time pad is considered a perfectly secure encryption scheme when used correctly, as it provides complete secrecy.
10. True or False: The access matrix approach, if stored directly, can be large and cumbersome to manage effectively.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to iterate over elements repeating each as many times as its count.
12. Write a python function to find minimum sum of factors of a given number.

*End of Paper*